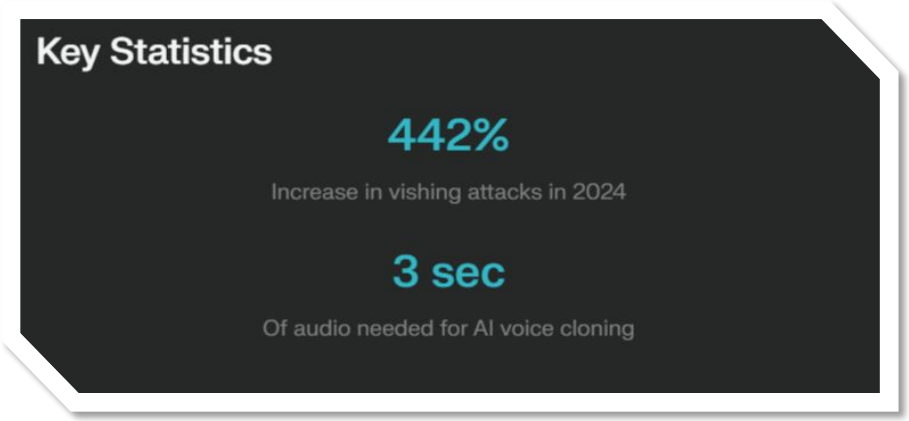
VishKill Pro: On-Device AI Defense Against Voice Phishing Scam

Institute Name: NIT Rourkela

Team Name: Phish ‘n Chips



Introduction

- Voice phishing or "vishing" represents a critical cybersecurity threat in today's digital landscape:
- Sophisticated AI-Generated Attacks: Modern scammers leverage advanced AI voice cloning technology to impersonate trusted contacts with unprecedented accuracy
- Real-time Detection Challenge: Traditional security measures fail against AI-powered voice synthesis that can replicate speech patterns, emotional tone, and vocal characteristics
- Our Solution: VishKill Pro provides a comprehensive HTTP API that combines multiple AI/ML modules for real-time audio analysis and scam detection
- Multi-layered Approach: Integrates deepfake detection, background noise analysis, speaker diarization, emotion recognition, and intelligent scam pattern analysis

Problem Domain: Voice Phishing Fraud

Vishing Attack Growth

AI-powered voice cloning makes attacks nearly indistinguishable from legitimate calls

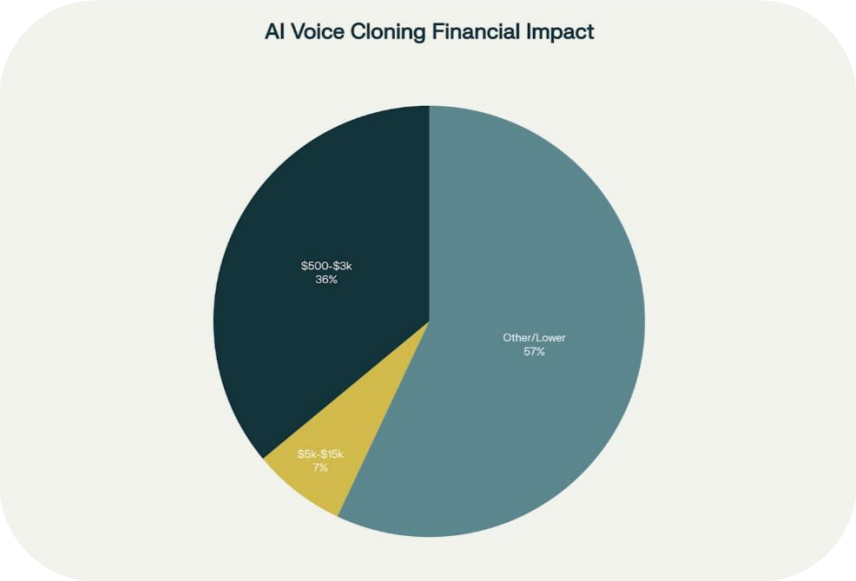
Background audio manipulation creates false environments (office noise, public spaces)

Emotional manipulation techniques exploit psychological vulnerabilities

Multi-speaker scenarios complicate traditional detection methods

Financial Impact

- 36% of victims lost between $500-$3,000
- 7% of victims lost between $5,000-$15,000



Problem Statement

The Challenge

Current vishing defense mechanisms are fundamentally inadequate against sophisticated AI-powered attacks that can bypass traditional security measures and exploit human psychology in real-time.

The Challenge & Market Gap

Critical Needs

- Multi-modal detection combining voice authentication, background analysis, and behavioral patterns
- High-accuracy classification (>95%) across multiple fraud vectors
- Scalable API solution that can integrate with existing security infrastructure
- Real-time processing capabilities for live call analysis
- Privacy-preserving architecture with secure data handling

Current Solution Limitations

- Lack of real-time deepfake voice detection capabilities
- Insufficient behavioral pattern analysis during active calls
- Missing comprehensive audio environment assessment
- Absence of integrated multi-modal threat detection systems

Project Objectives

🔍 Advanced Detection Capabilities

- Achieve >80% accuracy in identifying AI-generated voice clones
- Real-time deepfake detection using state-of-the-art ML models
- Comprehensive background noise analysis and environment assessment

🛡️ Multi-layered Security Analysis

- Speaker diarization for multi-participant call analysis
- Emotion detection to identify manipulation patterns
- LLM-powered scam content analysis and threat scoring

⚡ High-Performance API Design

- Sub-second response times for real-time conversations.
- Scalable architecture supporting concurrent audio processing
- RESTful API with comprehensive error handling and logging

📊 Comprehensive Threat Intelligence

- Detailed analysis reports with confidence scores
- Behavioral pattern identification and risk assessment
- Structured JSON responses for automated decision-making

🔄 Adaptability & Integration

- Modular architecture allowing easy integration with existing systems
- Support for multiple audio formats and quality levels
- Extensible framework for adding new detection modules

👤 Usability

Create a seamless user experience with minimal friction and clear security alerts

Proposed Solution

Audio Preprocessing & Validation

- Format conversion and standardization (16kHz mono WAV)
- File validation and security checks
- Temporary storage with automatic cleanup

Parallel Analysis Engine (Async Processing)

- Deepfake Detection: HuggingFace as1605/Deepfake-audio-detection-V2
- Background Noise Analysis: YAMNet from TensorFlow Hub
- Speaker Diarization: Pyannote-audio 3.1 pipeline
- Speech Transcription: OpenAI Whisper (medium model)

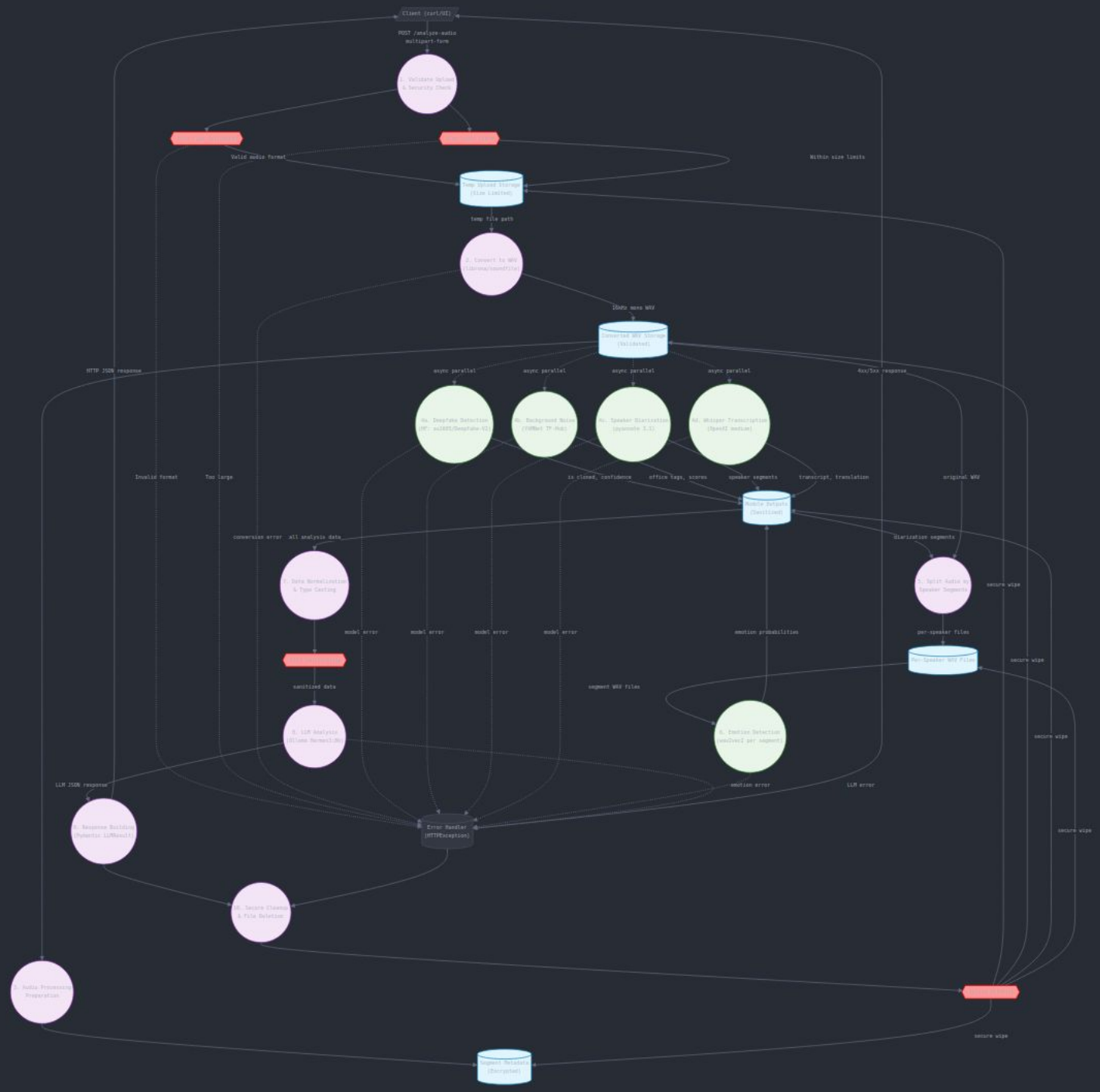
Sequential Processing Pipeline:

- Audio segmentation by speaker identity
- Per-segment emotion analysis using Wav2Vec2
- Data normalization and type casting

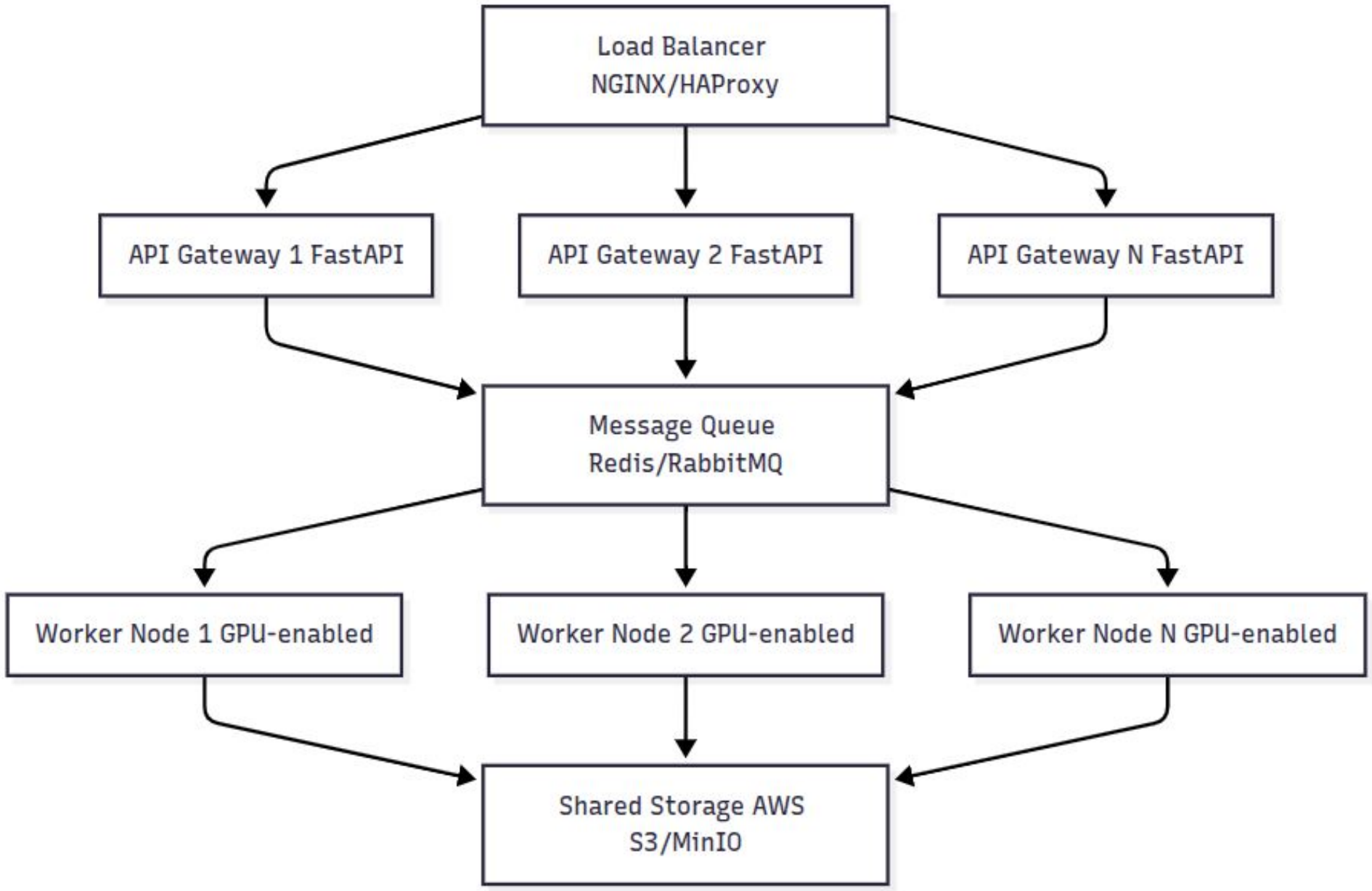
Intelligent Threat Assessment:

- LLM-powered analysis using Ollama Hermes3:8b
- Contextual scam pattern detection
- Risk scoring and decision reasoning

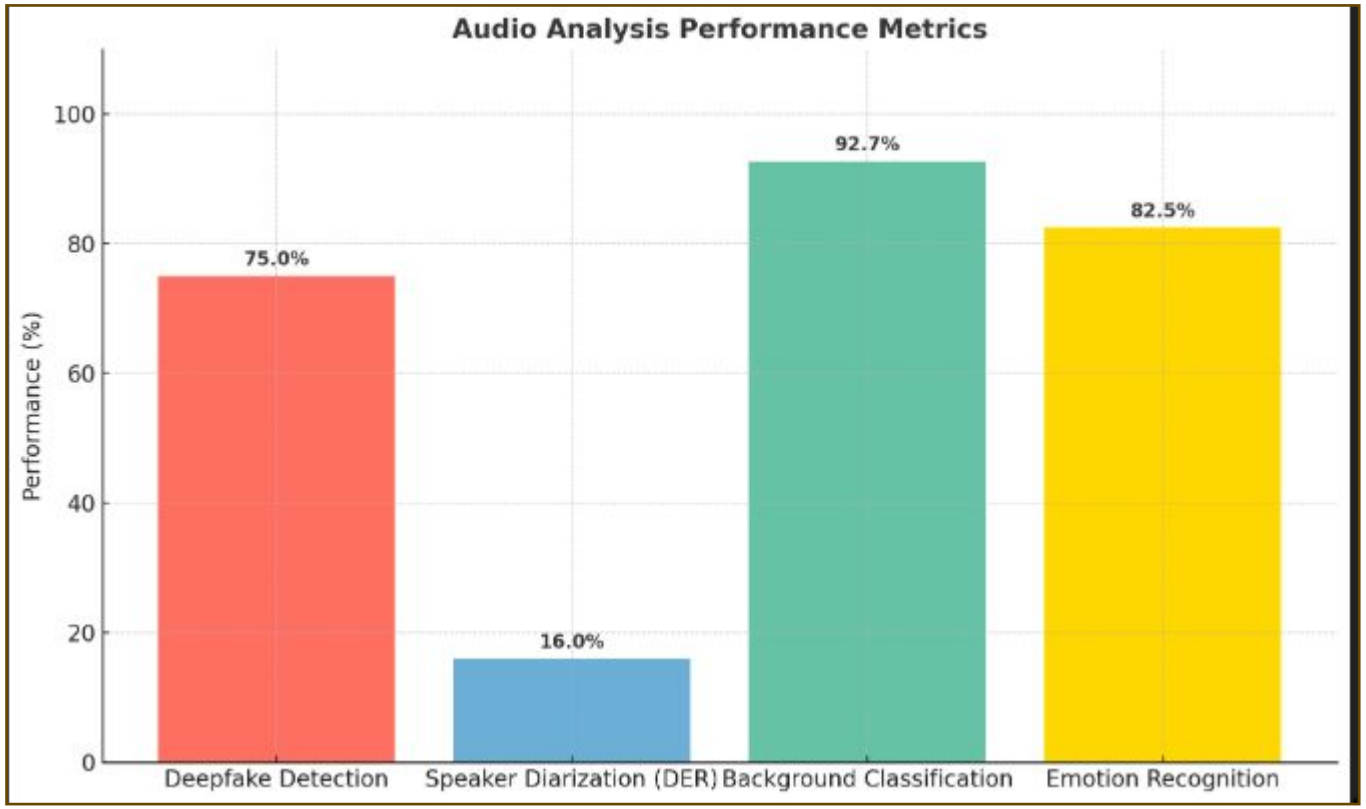
Level 1 DFD : VishKill



VishKill Scaling



Result Outcome



- Performance Metrics
- Audio Deepfake Detection: ~70–80% accuracy for detecting AI-generated voices
- Speaker Diarization (Pyannote): Diarization Error Rate (DER) ~12–20% (lower is better)
- Background Sound Classification (YAMNet): ~92.7% classification accuracy
- Emotion Recognition (Wav2Vec2): ~74–91% accuracy across common emotional categories
- Processing Time: <12 seconds for 15-second audio clips
- API Response Time: Sub-second latency for status endpoints

Work Completion Status

Completed

- System Architecture Design: Complete API specification and data flow
- Core ML Pipeline Development: All detection modules implemented and tested
- Audio Processing Framework: Conversion, validation, and preprocessing complete
- Security Implementation: Input validation, cleanup, and error handling
- Initial Testing: Unit tests and integration tests for all modules

In Progress

- Performance Optimization: GPU utilization and memory management
- Advanced Threat Modeling: Enhanced LLM prompt engineering
- API Documentation: Comprehensive OpenAPI specification
- Container Orchestration: Kubernetes deployment configurations

Planned

- Production Deployment: Cloud infrastructure setup and scaling
- Monitoring & Analytics: Performance metrics and usage analytics
- Client SDK Development: Python and JavaScript client libraries
- Advanced Features: Real-time streaming analysis and webhook integrations

Overall Project Completion : 75%

Project Timeline

Mar-Apr 2025
System architecture, technology selection

Apr-May 2025
ML pipeline, audio processing modules

May-Jun 2025
FastAPI implementation, endpoint testing

Jun-Jul 2025
Performance tuning, security hardening

Jul 2025
Production-ready prototype, demo preparation

July–Oct. 2025
Beta Testing
UI UX finetuning

Nov. 2025
Public Release Dec 2025 Planned

References:

- Deepfake Audio Detection:
  - as1605/Deepfake-audio-detection-V2 (HuggingFace model repository)
  - Generalized Source Tracing: Detecting Novel Audio Deepfake Algorithm with Real Emphasis and Fake Dispersion Strategy. arXiv:2406.03240
  - Deepfake Audio Detection Using Spectrogram-based Feature and Ensemble of Deep Learning Models. arXiv:2407.01777
- Background Noise Analysis:
  - YAMNet from TensorFlow Hub
- Speaker Diarization:
  - Pyannote-audio 3.1 pipeline
- Speech Transcription:
  - OpenAI Whisper (medium model)
- Emotion Recognition:
  - Wav2Vec2-based models for audio emotion analysis
- LLM for Threat Assessment:
  - Ollama Hermes3:8b model for contextual scam pattern detection
- Voice Phishing Detection with Small Language Models:
  - Detecting Voice Phishing with Precision: Fine-Tuning Small Language Models. arXiv:2506.06180

Product and Team

VishKill Pro :

- GitHub Repository: https://github.com/anivenk25/Phish-Chips.git
- System Design Diagram: https://claude.ai/public/artifacts/8f8be679-7d8c-45f0-8c50-6c75b4c9751b

Team Phish 'n Chips

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